VisionEvac: CPU-Bound Pedestrian Tracking and Bottleneck Calibration using YOLOv8 and ByteTrack

Osam Sami
COMSATS University Islamabad, Wah Campus
Wah Cantt, Pakistan
osamsamiawan0465@gmail.com
ORCID: https://orcid.org/0009-0001-3566-120X

Muhammad Haris Khan
COMSATS University Islamabad, Wah Campus
Wah Cantt, Pakistan
hariskhan803@gmail.com

***Abstract*—Crowd bottlenecks such as doors and gates limit the capacity of a building, and evacuation simulations that test them need measured pedestrian parameters. Measuring these from video on low-cost hardware is difficult, because inference must run on a CPU and people seen from above are small, tightly packed and partly occluded. This paper tests how far a pretrained YOLOv8 detector with the ByteTrack tracker can go under these limits, without fine-tuning, on top-view footage of a 2021 Jülich bottleneck experiment. We compare YOLOv8-nano and YOLOv8-small at 416×416 and 640×640 input on a laptop CPU. YOLOv8-small at 416×416 is the fastest setting (10.72 FPS) and reaches a tracking consistency of 45.6% in a 60-frame window, while YOLOv8-nano at the same size creates no track at all, since every detection score stays below the tracker's initiation threshold. On a 250-frame reference evaluation, however, the 416×416 setting is inaccurate (MOTA -6.63%, IDF1 5.26%, flow MAE 4.50 pedestrians per second), and the 640×640 setting, at 5.77 FPS, is only moderately better (MOTA 4.78%, IDF1 28.94%, flow MAE 0.80). Almost all pedestrians in the approach queue are missed. We report these figures as limits of zero-shot edge inference. The pipeline runs on an edge-class CPU and gives a baseline, but calibration of simulation models from its output will need a detector adapted to top-view crowds.**

***Index Terms*—pedestrian dynamics, bottleneck flow, multi-object tracking, YOLOv8, ByteTrack, CPU inference, model calibration**

I. Introduction

Bottlenecks such as doors, gates and narrow corridors decide how many people can leave or enter a facility per unit of time, and their width is among the first things a planner has to fix. Simulation models of pedestrian dynamics are therefore used to compare door widths and exit positions before construction. The social force model of Helbing et al. [4] showed that jamming and clogging at exits can be reproduced in simulation, and it is still the starting point for many later models.

The value of such a simulation depends on its parameters. Quantities like the desired walking speed or the minimum spacing between people are frequently copied from textbooks, and not measured in the situation that is modelled. Johansson et al. [5] adjusted the parameters of the social force model to video tracking data with an evolutionary algorithm, and Seer et al. [6] used several Kinect sensors to obtain trajectories for the calibration of three model variants. Both works indicate that measured data improve the model, but both rely on special equipment or on a heavy optimisation step.

Good empirical data on bottleneck flow do exist. Kretz et al. [8] measured fluxes and time gaps for ten bottleneck widths, Seyfried et al. [9] reported that the flow grows linearly with the width and that a jam can occur even if the incoming flow stays below the nominal capacity, and Adrian et al. [10] studied how corridor width, motivation and social norms change the crowd in front of an entrance. The trajectories behind these studies were mostly produced with dedicated software such as PeTrack [2], which gives very accurate results but expects an experiment that was prepared with markers and a suitable camera setup.

General-purpose detection and tracking has become fast and openly available in the meantime. ByteTrack [1] keeps low-confidence detection boxes and associates them with existing tracklets, which reduces missed objects and broken trajectories, but it was evaluated on general benchmarks such as MOT17 and MOT20. Head-detection benchmarks for dense crowds, for example RPEE-Heads [3], report that the best detectors reach roughly 90 percent mean average precision, although this number concerns detection alone and says nothing about trajectories, flow or CPU speed. To our knowledge there is little evidence on what an ordinary laptop CPU, a pretrained YOLOv8 detector [13] and ByteTrack can deliver on top-view bottleneck video, where heads and shoulders are small and partly hidden by neighbours. This is the gap that the present paper addresses.

We therefore built VisionEvac, a pipeline that detects and tracks pedestrians with YOLOv8 and ByteTrack on a CPU, converts pixel positions to metres, and derives density and flow, which are the quantities an agent-based model would be calibrated against. We test it on the 2021 bottleneck experiment of Forschungszentrum Jülich [14] without fine-tuning the detector. The outcome is mixed. Throughput on a CPU is usable, but tracking accuracy is low, and the setting that is fastest is also the least accurate. We report both sides,

and we trace the main failures to measurable causes.

The contributions of this paper are:

- A CPU-only pipeline that links YOLOv8, ByteTrack and a pixel-to-metre conversion to the estimation of pedestrian density and flow at a bottleneck, applied without fine-tuning to top-view Jülich footage and run on an Intel Core i5-8250U laptop processor.
- A four-configuration ablation over model size (nano, small) and input size (416, 640), together with a diagnosis showing that YOLOv8-nano at 416×416 produces no track because its maximum detection score (0.2177) stays below the ByteTrack initiation threshold (0.25).
- An honest reference evaluation with CLEAR MOT metrics, IDF1, density error and flow error for the 416×416 and 640×640 small models (best MOTA 4.78%, best IDF1 28.94%, best flow MAE 0.80), and a measurement showing that nearly all pedestrians in the approach queue are not detected at the operating threshold.

II. Related Work

A. Detection and trajectory extraction

ByteTrack [1] is a tracking-by-detection method. Instead of discarding detections with a low score, it uses them in a second association step, which helps to recover occluded people and to reduce fragmented trajectories. YOLOv8 [13] supplies the detections in our work and is available in several sizes, of which we use the nano and small variants. For crowd video, the RPEE-Heads benchmark [3] offers about 110 thousand annotated heads from 66 videos of railway platforms and event entrances. In pedestrian research, PeTrack [2] extracts accurate trajectories from video of marked pedestrians, at the price of a prepared experiment.

B. Pedestrian models and calibration

Helbing et al. [4] studied escape panic with a force-based model and showed that simulation can expose the mechanisms behind jamming. Johansson et al. [5] specified the social force model by fitting it to video tracking data, and Seer et al. [6] calibrated model variants with Kinect trajectories, with a reported pedestrian detection rate of up to 94 percent. Rzezonka et al. [7] used a simple velocity-based model to test whether a physical mechanism can explain the dependence of the density in front of a bottleneck on the corridor width. None of these works targets a lightweight, CPU-only extraction of the measurements.

C. Bottleneck experiments

Kretz et al. [8] found that the specific flux declines linearly with the width as long as only one person can pass at a time, and stays constant for wider bottlenecks. Seyfried et al. [9] compared several experiments and concluded that jams can appear below the capacity derived from the fundamental diagram. Adrian et al. [10] added the role of motivation and of a primed queuing norm. These studies provide the reference data, but they do not describe an automated video pipeline.

D. Evaluation metrics

The CLEAR MOT metrics of Bernardin and Stiefelhagen [11] summarise tracking accuracy through MOTA, which combines false positives, misses and identity switches, and MOTP, which measures localisation error. IDF1 [12] instead measures how long identities are preserved. We report both families, because they react differently to fragmented tracks.

III. Methodology

A. Data and hardware

We use the clip `4D010_w120_l200_h0` of the 2021 bottleneck experiment published by Forschungszentrum Jülich [14] under a CC BY 4.0 licence, together with its PeTrack reference trajectories. The video has a resolution of 4000×3000 pixels at 25 frames per second and 1889 frames; the reference file holds 136 pedestrians. The corridor is 2 m long and 1.2 m wide in the plots ($|y| \leq 0.6$ m). Two frame windows are used: frames 350 to 409 (60 frames, 2.4 s) for the ablation grid, and frames 300 to 549 (250 frames, 10 s) for the reference evaluation. In the latter window the reference file contains 116 pedestrians in the whole scene. No part of the detector was fine-tuned on this material, so all results are zero-shot with respect to the scene.

All timings were measured on an Intel Core i5-8250U (4 cores, 8 logical processors, 1.60 GHz) with 7.86 GB of RAM under Windows 10, using Python 3.12.10, Ultralytics 8.4.191 and PyTorch 2.15.0 (CPU build). The reported FPS includes pre-processing (resizing and normalisation), the network forward pass, non-maximum suppression and the ByteTrack association. It excludes video decoding, because the frames were held in memory beforehand, as well as video writing, logging and plotting. Each configuration was timed in a single run, after one unmetered warm-up pass.

B. Pipeline

The pipeline has four steps. First, a pretrained YOLOv8 model [13] (class *person*) processes each full frame, which is resized to the chosen square input size; the detector confidence threshold is 0.08 and the NMS IoU threshold 0.70. Second, ByteTrack [1] associates the detections over time with the default Ultralytics settings: a high-score threshold of 0.25, a low-score threshold of 0.10, a new-track threshold of 0.25, a track buffer of 30 frames and a matching threshold of 0.80. In this implementation a newly created track is output only after it has been matched in the following frame. Third, pixel positions are converted to metric floor coordinates by a linear mapping with a scale of 272 pixels per metre, $x_m = (625 - y_{px})/272$ and $y_m = (2035 - x_{px})/272$, so that the entrance lies at $x = 0$ m and the exit at $x = 2$ m. The mapping is a plain linear scale and contains no correction of the lens distortion that is visible in the footage. Fourth, density and flow are computed from the metric positions and compared with the same quantities computed from the reference file.

TABLE I

ABLATION OF MODEL SIZE AND INPUT RESOLUTION ON A CPU, FRAMES 350–409 ($T = 60$). CONS. IS THE TRACKING CONSISTENCY OF EQ. (1).

Model	Input	FPS	Det.	Tracks	Avg. len.	Cons. (%)
YOLOv8-nano	416	8.48	61	0	0.0	0.0
YOLOv8-nano	640	6.79	99	3	4.7	7.8
YOLOv8-small	416	**10.72**	100	3	**27.3**	**45.6**
YOLOv8-small	640	5.77	174	12	14.5	24.2

C. Evaluation protocol

Tracking consistency. For a window of T frames, let U be the set of track identities output by ByteTrack and L_i the number of frames in which identity i is output. The average track length is $\bar{L} = \frac{1}{|U|} \sum_{i \in U} L_i$ (zero if U is empty), and the tracking consistency is

$$C = \frac{\bar{L}}{T} \times 100\%. \tag{1}$$

A value of 0% therefore means that no track was ever output. The measure is a persistence score, not a measure of identity correctness, and it can be high for tracks that follow the wrong person. For the ablation grid, $T = 60$.

Reference evaluation. Tracking is compared with the reference trajectories through MOTA, MOTP, IDF1, identity switches, false positives and misses, restricted to the zone $-0.5 \le x \le 2.2$ m. In frames 300 to 549 this zone contains 32 reference pedestrians and 2908 reference point observations. Because the zone starts at $x = -0.5$ m, it excludes most of the approach queue, so the MOT metrics describe the corridor and its entrance only. Flow is the number of pedestrians crossing the line $x = 1.0$ m in one-second bins, and density is the mean pedestrian density in the corridor; both are compared with the reference by the mean absolute error (MAE).

IV. EXPERIMENTS AND RESULTS

A. Ablation over model size and input resolution

Table I and Fig. 1 give the four configurations over the 60-frame window. Inference time per frame is the reciprocal of the speed; for the fastest setting, 93.2 ms corresponds to 10.72 FPS (latencies of the two small-model settings are listed in Table III).

Two points need care in reading this table. The 416 setting with the small model has the highest speed and the longest average track, but its consistency of 45.6% rests on only three tracks and says nothing about whether they follow the right people. In addition, YOLOv8-nano at 416 ran slower (8.48 FPS) than YOLOv8-small at 416 (10.72 FPS), which is the opposite of what the model sizes suggest. We timed each configuration once on a laptop CPU and cannot say how much of the difference is measurement noise, so we do not interpret it further.

B. Why YOLOv8-nano at 416×416 produced no track

The diagnosis in Table II was run on the same 60 frames. ByteTrack creates a new track only for a detection whose score

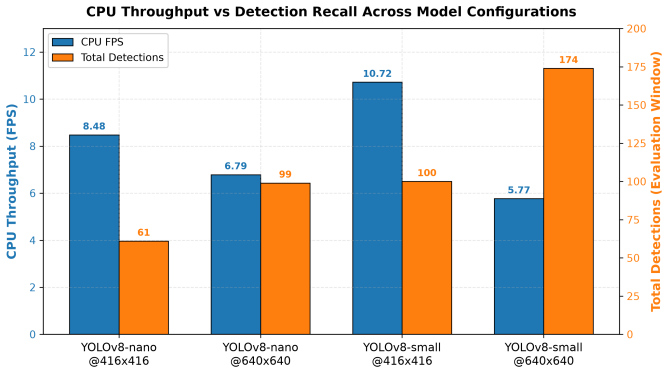


Fig. 1. CPU throughput (left axis, FPS) and total number of detections in the 60-frame evaluation window (right axis) for the four model configurations. Raising the input size from 416 to 640 increases the number of detections for both models (61 to 99 for nano, 100 to 174 for small) but lowers the throughput (8.48 to 6.79 and 10.72 to 5.77 FPS). The detection count is only a proxy for recall, because the figure does not show whether a detection is correct.

TABLE II

DETECTION SCORES AGAINST THE BYTETRACK NEW-TRACK THRESHOLD OF 0.25 (FRAMES 350–409). BOX SIZE IS THE MEDIAN PERSON BOX AFTER RESIZING TO THE MODEL INPUT, IN PIXELS (WIDTH×HEIGHT); THE ORIGINAL BOXES ARE ABOUT 150–180×220–260 PX IN THE 4000×3000 FRAME.

Config.	Det.	Max	Median	≥ 0.25	Tracks	Box
nano 416	61	0.2177	0.1089	0	0	17.6×24.6
nano 416, c=0.05	153	0.2177	0.0687	0	0	17.8×26.6
nano 640	99	0.4639	0.1090	8	3	23.6×35.8
small 416	100	0.5290	0.2260	42	3	18.6×23.2
small 640	174	0.6545	0.2673	89	12	22.8×38.3

reaches the new-track threshold of 0.25. The largest score of YOLOv8-nano at 416 over all 60 frames was 0.2177, so none of its 61 detections could start a track, even though 29 of them overlapped the box of the previous frame with an IoU of at least 0.80. Lowering the detector threshold from 0.08 to 0.05 raised the detections from 61 to 153 but produced no track either, because the largest score remained 0.2177. The cause of the 0% consistency is therefore a score problem and not an association problem.

Why the nano model scores so low is a further question. The table shows that after the 4000×3000 frame is reduced to 416×416, a person is only about 18×25 pixels large, and the nano model has the least capacity to recognise such small objects. This agrees with the lower scores of nano at both sizes, but we did not test it in isolation, so it remains an explanation consistent with the data and not a proven cause.

Why YOLOv8-small. The small model has higher scores at the same input (median 0.2260 against 0.1089 at 416), and 42 of its 100 detections pass the new-track threshold. It was therefore the only model that gave usable tracks within the CPU budget, and 416 was the fastest setting. This choice is a choice for speed. The next subsection shows that it costs accuracy.

TABLE III
REFERENCE EVALUATION OF YOLOV8-SMALL ON CPU, FRAMES 300–549 (250 FRAMES), ZERO-SHOT. MOT METRICS REFER TO THE ZONE $-0.5 \leq x \leq 2.2$ M (2908 REFERENCE OBSERVATIONS).

Metric	Small, 416	Small, 640
CPU FPS	10.72	5.77
Latency (ms/frame)	93.2	173.3
Predicted boxes (250 frames)	503	990
Unique track IDs	20	43
MOTA (%)	−6.63	4.78
MOTP (m)	0.449	0.513
IDF1 (%)	5.26	28.94
Identity switches	32	10
False positives	268	417
Misses	2611	2342
Mean density, reference (ped/m^2)	2.07	2.05
Mean density, predicted (ped/m^2)	0.16	0.62
Density MAE (ped/m^2)	1.92	1.43
Mean flow, reference (ped/s)	2.10	2.10
Mean flow, predicted (ped/s)	4.20	1.30
Flow MAE (ped/s)	4.50	0.80

C. Reference evaluation: 416 against 640

Table III compares the two small-model settings on frames 300 to 549 with the same reference data.

The 416 setting is almost twice as fast but is clearly the less accurate one. Its MOTA is negative (-6.63%), because misses (2611) and false positives (268) together with 32 identity switches exceed the number of reference observations that are correctly matched. Its IDF1 of 5.26% means that identities are almost never preserved. The 640 setting is better on every accuracy measure except MOTP and the number of false positives, but its MOTA of 4.78% and IDF1 of 28.94% are still low. Its misses (2342) are far more frequent than false positives (417) and identity switches (10), so the detector fails mainly by not seeing people. The MOTP of 0.513 m (0.449 m for 416) shows that matched detections lie about half a metre from the reference, which is large against a corridor that is 1.2 m wide. We regard all these values as limits of zero-shot edge inference, and not as a competitive tracking result.

The flow error of the 416 setting has the opposite sign to the density error: the predicted flow (4.20 ped/s) is twice the reference (2.10 ped/s), while the predicted density (0.16 ped/m^2) is far below the reference (2.07 ped/m^2). A plausible reason is that broken tracks are counted as several separate crossings of the line $x = 1.0$ m, but we did not verify this. The 640 setting underestimates both quantities. Its flow MAE of 0.80 ped/s is the best value we obtained, but because the 640 setting is also the slower one, the best accuracy and the best speed come from different configurations, and no single setting is good at both.

D. Pedestrians in the approach zone are not detected

The MOT zone of Table III hides the largest failure. In frame 300 the reference contains 116 pedestrians in the scene, of which 110 stand in the approach zone ($x < 0$ m) and only 6 in the corridor; frames 400 and 500 show the same pattern (102 of 107 and 94 of 99). The saved predictions of the 640 run contain 990 boxes, of which only 5 (0.51%) lie in the approach zone, none of them in the three inspected frames. Fig. 2 shows these frames. The queue is fully inside the field of view, and the pedestrians are visible, but the detector gives them scores that are too low. When the confidence threshold was lowered to 0.01, the raw output of YOLOv8-small contained 30, 22 and 8 boxes in the approach zone of frames 300, 400 and 500, against 110, 102 and 94 reference pedestrians. In frames 400 and 500 the highest score among these boxes was 0.068 and 0.076, below the detector threshold of 0.08 and far below the tracker threshold of 0.25. Isolated pedestrians in the corridor reached scores up to 0.508.

The cause that we can state with evidence is the low confidence of the pretrained detector on this view. We think that the overhead viewpoint, the dense packing, and the caps with fiducial markers worn by the participants all make the appearance unusual for a model trained on COCO, but we did not separate these factors.

E. Spatial and temporal comparison with the reference

Fig. 3 compares the spatial density of the reference (PeTrack) with that of the 640 pipeline. The reference shows its highest densities in the approach zone, while the predicted map is almost empty there and contains only weak density inside the corridor and around the exit. Fig. 4 shows the flow per one-second bin: the estimate stays at 1 ped/s for the first six seconds while the reference is at 3 and then 2 ped/s, and it reaches the reference value in only three of the ten bins. Fig. 5 places predicted and reference trajectories in the same metric frame. The reference paths run continuously from the approach zone through the corridor, whereas the predicted paths are short fragments that start at about $x = 0.7$ m or later, and several of them lie outside the corridor band ($|y| > 0.6$ m).

In the exported prediction file of the 640 run, the 250 frames contain 990 boxes assigned to 43 track identities, so the average identity lives for about 23 frames, and a frame holds on average 3.96 tracked persons (maximum 8). One identity persists for all 250 frames while moving by less than 0.15 m; in the audit frames it belongs to a person standing on the lifting platform at the left of the corridor, so it is a real person who does not take part in the flow.

V. DISCUSSION AND LIMITATIONS

Four limitations should be kept in mind. First, the detector was used zero-shot. A model pretrained on general images sees dense top-view crowds rarely, and fine-tuning on head-annotated crowd data such as RPEE-Heads [3], or lowering the tracker thresholds together with a better detector, would be the obvious next steps; neither was tested here. Second, the ablation grid covers 60 frames and each configuration was timed once, while the reference evaluation covers 250 frames of the same clip; all results come from one clip, one camera view and organised walking, so they do not support statements about panic or safety. Third, the MOT metrics are

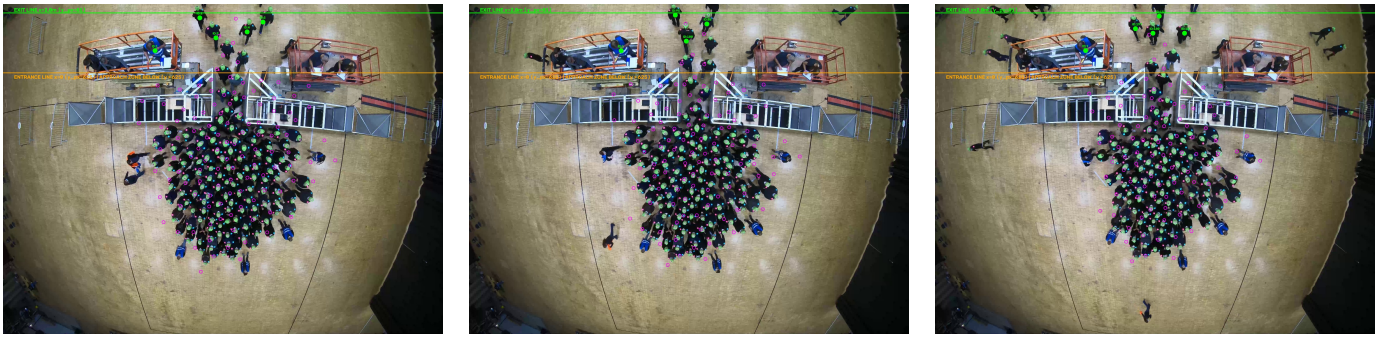


Fig. 2. Audit frames 300 (left), 400 (centre) and 500 (right) of the top-view video. Magenta circles mark the reference pedestrian positions projected into the image, green filled circles with identity labels mark the tracked detections of the YOLOv8-small and ByteTrack output, and the horizontal lines mark the exit ($x = 2$ m, top) and the entrance ($x = 0$ m). The dense queue below the entrance line contains almost all pedestrians of the scene, but it receives almost no tracked detection; the detections are concentrated above the entrance line. The overlays also show a displacement of some projected reference positions from the heads, which reflects the strong lens distortion of the footage that the linear mapping does not correct.

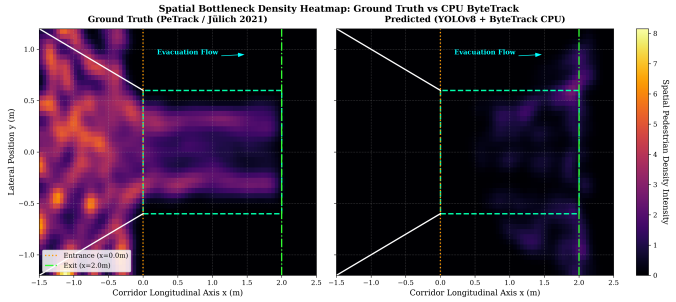


Fig. 3. Spatial pedestrian density at the bottleneck in metric corridor coordinates: reference trajectories from PeTrack (left) and the YOLOv8-small and ByteTrack output on CPU at 640×640 input (right). Vertical lines mark the entrance ($x = 0$ m) and the exit ($x = 2$ m), white lines the walls of the approach funnel, and dashed lines the corridor edges at $y = \pm 0.6$ m; the colour scale is the one produced by our plotting script. The reference concentrates density in the approach zone ($x < 0$ m), whereas the predicted map is nearly empty there and shows only weak density inside the corridor and near the exit. This pattern is consistent with the underestimated mean density of 0.62 ped/m^2 against 2.05 ped/m^2.

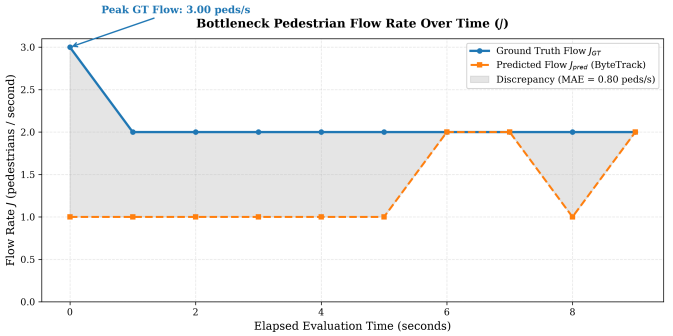


Fig. 4. Flow rate J over elapsed time in one-second bins for the reference (J_{GT}) and the 640×640 pipeline (J_{pred}); the shaded area marks the discrepancy. The reference peaks at 3 ped/s in the first bin and stays at 2 ped/s afterwards, while the prediction remains at 1 ped/s for the first six seconds, matches the reference at 6, 7 and 9 s, and drops to 1 ped/s at 8 s. The mean values are 2.1 ped/s (reference) and 1.3 ped/s (prediction), which gives the flow MAE of 0.80 ped/s.

computed in the zone $-0.5 \leq x \leq 2.2$ m and therefore do not measure the large number of missed pedestrians in the queue; the real tracking recall of the scene is lower than the corridor metrics suggest. Fourth, the pixel-to-metre mapping is linear and ignores lens distortion, which adds position error away from the image centre and may contribute to the MOTP values.

The agent-based calibration stage, in which measured speed and spacing would be fed into a Mesa model of the bottleneck, has not been evaluated and is left for future work. With a MOTA below 5% at best, the present measurements are not accurate enough to calibrate such a model, and we do not claim that they are.

VI. Conclusion

We examined whether a pretrained YOLOv8 detector with ByteTrack can supply density and flow measurements for bottleneck analysis on a CPU. YOLOv8-small at 416×416 ran at 10.72 FPS, but its reference accuracy was poor (MOTA -6.63%, IDF1 5.26%, flow MAE 4.50 ped/s). The 640×640 setting gave the best accuracy we measured (MOTA 4.78%, IDF1 28.94%, flow MAE 0.80 ped/s) at 5.77 FPS, which is still low. YOLOv8-nano at 416×416 produced no track, because its largest score (0.2177) stayed below the tracker's initiation threshold (0.25), and the queue in front of the entrance was almost never detected by any setting. The pipeline is therefore a working and reproducible CPU baseline and not yet an accurate measurement tool. Fine-tuning on top-view crowd data, correction of the lens distortion, and the calibration of an agent-based model are the next steps.

References

[1] Y. Zhang, P. Sun, Y. Jiang, D. Yu, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-object tracking by associating every detection box," arXiv:2110.06864, 2021.

[2] M. Boltes and A. Seyfried, "Collecting pedestrian trajectories," *Neurocomputing*, vol. 100, pp. 127–133, 2013.

[3] M. Abubaker, Z. Alsadder, H. Abdelhaq, M. Boltes, and A. Alia, "RPEE-Heads: A novel benchmark for pedestrian head detection in crowd videos," arXiv:2411.18164, 2024.

[4] D. Helbing, I. Farkas, and T. Vicsek, "Simulating dynamical features of escape panic," *Nature*, vol. 407, pp. 487–490, 2000.

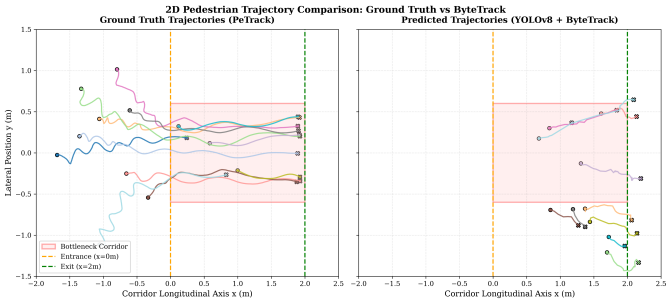


Fig. 5. Two-dimensional trajectories in metric corridor coordinates, reference (PeTrack, left) and ByteTrack on YOLOv8-small detections at 640×640 input (right). Circles mark the start and crosses the end of each trajectory; the shaded band is the corridor between the entrance ($x = 0$ m) and the exit ($x = 2$ m). Reference paths are continuous from the approach zone through the corridor. Predicted paths are short fragments confined mostly to $x > 0.7$ m, with no trajectory in the approach zone, and several lie outside the corridor band. This fragmentation corresponds to the low consistency and IDF1 values in Tables I and III.

[5] A. Johansson, D. Helbing, and P. K. Shukla, "Specification of the social force pedestrian model by evolutionary adjustment to video tracking data," *Advances in Complex Systems*, vol. 10, no. S2, pp. 271–288, 2007.

[6] S. Seer, N. Brändle, and C. Ratti, "Kinects and human kinetics: A new approach for studying crowd behavior," arXiv:1210.2838, 2012.

[7] J. Rzezonka, M. Chraibi, A. Seyfried, B. Hein, and A. Schadschneider, "An attempt to distinguish physical and socio-psychological influences on pedestrian bottleneck," *Royal Society Open Science*, vol. 9, no. 6, 2022.

[8] T. Kretz, A. Grünebohm, and M. Schreckenberg, "Experimental study of pedestrian flow through a bottleneck," *J. Stat. Mech.: Theory Exp.*, P10014, 2006.

[9] A. Seyfried, O. Passon, B. Steffen, M. Boltes, T. Rupprecht, and W. Klingsch, "New insights into pedestrian flow through bottlenecks," *Transportation Science*, vol. 43, no. 3, pp. 395–406, 2009.

[10] J. Adrian, A. Seyfried, and A. Sieben, "Crowds in front of bottlenecks at entrances from the perspective of physics and social psychology," *J. R. Soc. Interface*, vol. 17, no. 165, 20190871, 2020.

[11] K. Bernardin and R. Stiefelhagen, "Evaluating multiple object tracking performance: The CLEAR MOT metrics," *EURASIP J. Image Video Process.*, vol. 2008, Art. 246309, 2008.

[12] E. Ristani, F. Solera, R. Zou, R. Cucchiara, and C. Tomasi, "Performance measures and a data set for multi-target, multi-camera tracking," in *Proc. ECCV Workshops*, 2016.

[13] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," software, version 8.0.0, 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[14] Forschungszentrum Jülich, "Pedestrian Dynamics Data Archive: 2021 bottleneck experiment (varying motivation)," DOI 10.34735/ped.2021.9. [Online]. Available: https://ped.fz-juelich.de/da/2021bottleneck